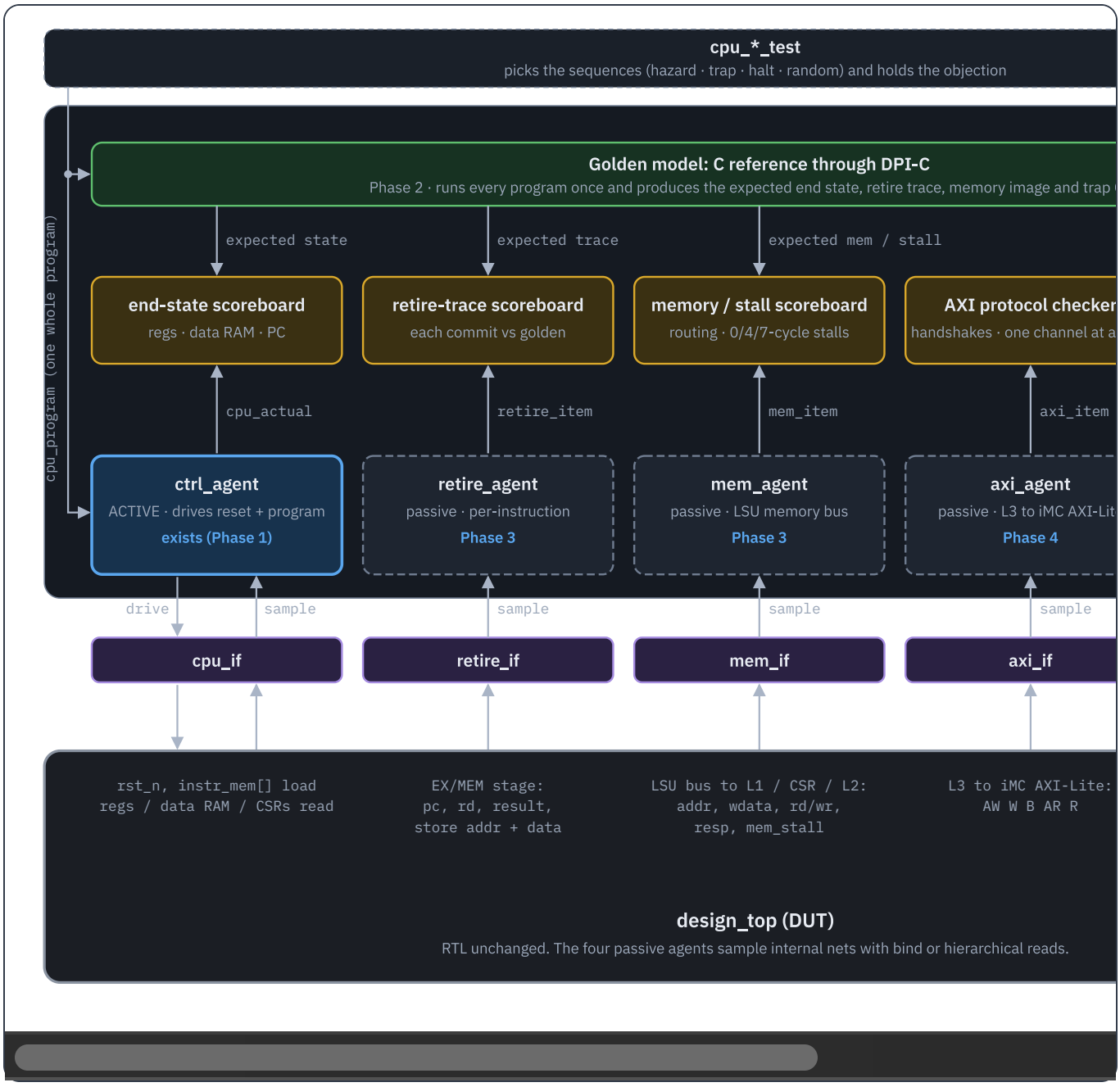
Multi-agent UVM plan for the RV32I CPU

Yes, it fits. The DUT has one real input (reset + program) but several independent behaviours worth watching, so each gets its own agent and its own checker.



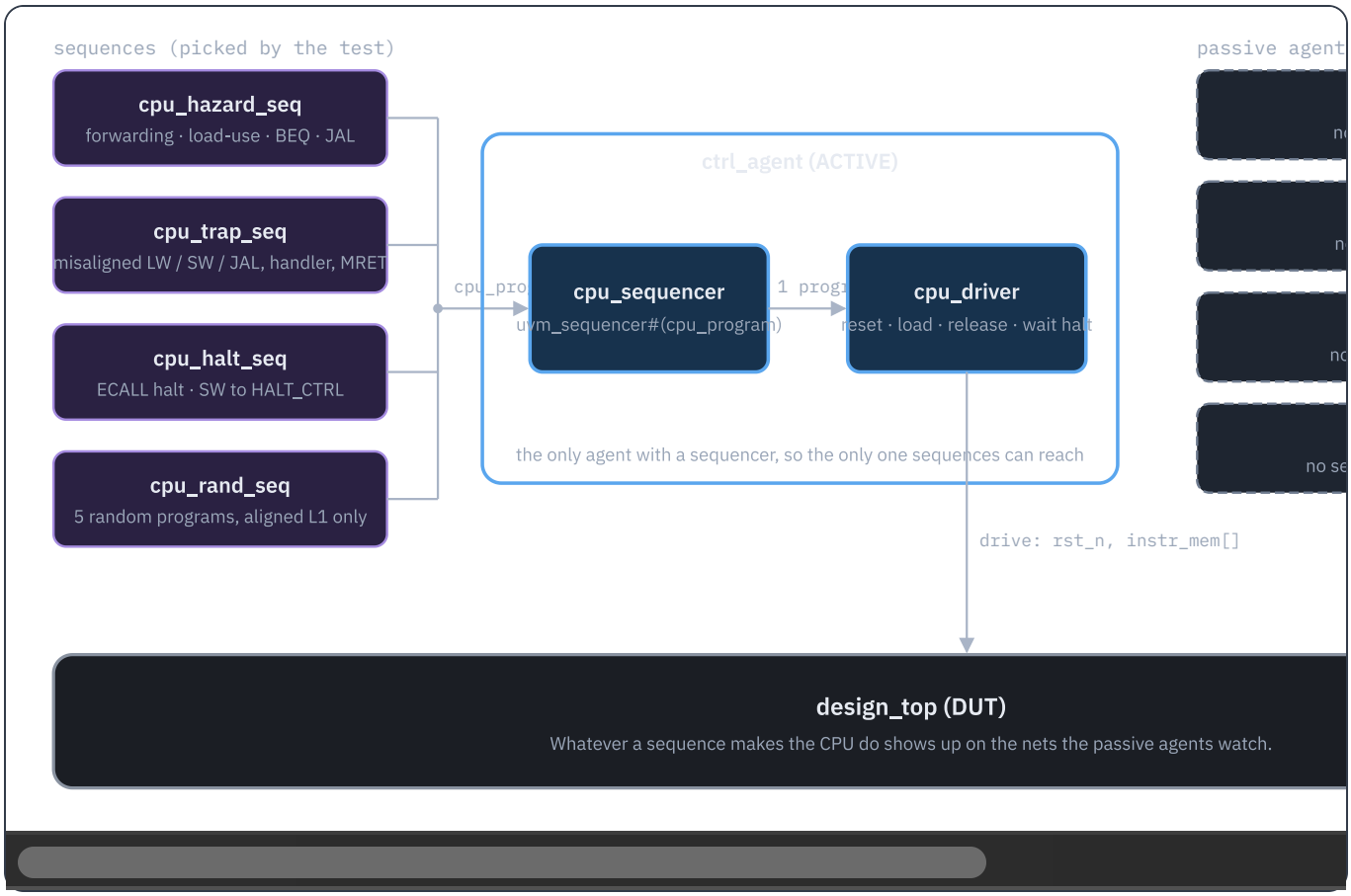
Solid blue = active agent that drives the DUT. Dashed grey = passive agent that only samples. Yellow = checker, green = golden model (Phase 2). Every agent publishes its own item type to its own checker.

THE FIVE AGENTS

AGENT	MODE	WATCHES	ITEM IT PUBLISHES	ENABLES
ctrl_agent	Active	reset, program load	cpu_actual	End-state compare vs

ctrl_agent	Active	reset, program load, end-state read-back	cpu_actual	End-state compare vs golden
retire_agent	Passive	EX/MEM: pc, rd, result, store addr/data	retire item per instruction	Locates the first wrong instruction, not just a wrong final state
mem_agent	Passive	LSU bus + mem_stall	mem item per access	Routing (L1/CSR/L2/L3) and 0/4/7-cycle stall checks
axi_agent	Passive now, active later	AW/W/B/AR/R between L3 and iMC	AXI transaction	Protocol checker, ready for a real AXI slave later
status_agent	Passive	halt, pc_out, fault flags, trap_taken	status event	Trap cause, PC freeze on halt, squash checks

SEQUENCES AND AGENTS



Sequences never connect to a passive agent. A sequence only produces `cpu_program` items for the one sequencer in `ctrl_agent`; the passive agents just report what those programs did to the DUT.

WHICH SEQUENCE LIGHTS UP WHICH AGENT

SEQUENCE	WHAT IT SENDS	RETIRE	MEM	AXI	STATUS	WHY IT EXISTS
`cpu_hazard_seq`	Forwarding, load-use stall, taken BEQ, JAL, LUI, x0	●	●	–	●	Checks the pipeline co the right values; the re sees each result
`cpu_trap_seq`	Misaligned LW, SW and JAL, then handler and MRET	●	●	–	●	Traps are only visible c status flags, and the fa instruction must not re write
`cpu_halt_seq`	ECALL halt; external store then SW to HALT_CTRL	●	●	●	●	PC must freeze and yo instructions squash; th store crosses the AXI
`cpu_rand_seq`	5 programs of 24 random ops, aligned L1 addresses	●	●	–	–	Broad coverage of op r it never leaves L1, so r AXI traffic

Gap: only `cpu_halt_seq` touches the external path, and only with one store. `mem_agent` and `axi_agent` need a new `cpu_mem_seq` (L2 hit, L3 hit, L3 miss, CSR access, out-of-range) before their checkers can prove the 0, 4 and 7-cycle stalls.

HONEST CAVEATS

The DUT exposes no bus ports, so the four passive agents are white-box probes (bind or hierarchical reads of internal nets). That is normal for a block-level DV project and still shows agent/monitor/scoreboard structure. A truly active `axi_agent` would need the iMC moved outside `design_top`, which is an RTL change, so it stays passive until you choose to do that.

SUGGESTED ORDER

Finish Phase 2 first (golden model through DPI-C). Then add `status_agent` and